Envelope Equations

USPAS - Beam Physics with Intense Space Charge

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Envelope equations: reduced-order characterization of beam dynamics in terms of second-order moments

[Image]

Road map

- Start from single particle equation of motion: $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
- Make use of:
  - Paraxial (near-axis) approximation
  - Conservation of angular momentum
  - Axisymmetry of external forces
- Arrive at single-particle paraxial ray equation.
- Take statistical averages to obtain moment/envelope equations.

Paraxial ray equation

Equation of motion in cylindrical coordinates

Equation of motion for particle of mass m and charge q in electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$:

$$\frac{d\mathbf{p}}{dt} = \frac{d(\gamma m \dot{\mathbf{x}})}{dt} = q\left(\mathbf{E} + \dot{\mathbf{x}} \times \mathbf{B}\right), \tag{1}$$

Switch to cylindrical coordinates (r, θ, z):

$$\begin{aligned} \mathbf{p} &= p_x \hat{x} + p_y \hat{y} + p_z \hat{z} \\ &= p_r \hat{r} + p_\theta^* \hat{\theta} + p_z \hat{z}. \end{aligned} \tag{2}$$

The components p_r and p_θ^* are given by

$$\begin{aligned} p_r &= \gamma m \dot{r}, \\ p_\theta^* &= \gamma m r \dot{\theta}. \end{aligned} \tag{3}$$

The symbol p_θ^* is used to reserve p_θ for the *canonical* angular momentum, defined later.

In cylindrical coordinates, the unit vectors depend on the particle position and thus change with time:

$$\begin{aligned}
\dot{\hat{r}} &\equiv d\hat{r}/dt = \dot{\theta}\hat{\theta}, \\
\dot{\hat{\theta}} &\equiv d\hat{\theta}/dt = -\dot{\theta}\hat{r}.
\end{aligned} \tag{4}$$

Thus, the time derivative of the momentum vector in Eq. (2) is:

$$\begin{aligned}
\frac{d\mathbf{p}}{dt} &= \frac{d}{dt}\left(p_r\hat{r} + p_\theta^*\hat{\theta} + p_z\hat{z}\right) \\
&= \dot{p}_r\hat{r} + p_r\dot{\hat{r}} + \dot{p}_\theta\hat{\theta} + p_\theta^*\dot{\hat{\theta}} + \dot{p}_z\hat{z} \\
&= \left(\dot{p}_r - p_\theta^*\dot{\theta}\right)\hat{r} + \left(\dot{p}_\theta + p_r\dot{\theta}\right)\hat{\theta} + \dot{p}_z\hat{z} \\
&= \left[\frac{d(\gamma m\dot{r})}{dt} - \gamma m r\dot{\theta}^2\right]\hat{r} + \left[\frac{d(\gamma m r\dot{\theta})}{dt} + \gamma m\dot{r}\dot{\theta}\right]\hat{\theta} + \left[\frac{d(\gamma m\dot{z})}{dt}\right]\hat{z}. \\
&= \left[\frac{d(\gamma m\dot{r})}{dt} - \gamma m r\dot{\theta}^2\right]\hat{r} + \left[\frac{1}{r}\frac{d}{dt}\left(\gamma m r^2\dot{\theta}\right)\right]\hat{\theta} + \left[\frac{d(\gamma m\dot{z})}{dt}\right]\hat{z}.
\end{aligned} \tag{5}$$

This takes care of the left side of the Lorentz force equation (Eq. (1)).

The right side of Eq. (1) requires the cylindrical field components:

$$\begin{aligned}\mathbf{E} &= E_r\hat{r} + E_\theta\hat{\theta} + E_z\hat{z},\\ \mathbf{B} &= B_r\hat{r} + B_\theta\hat{\theta} + B_z\hat{z}.\end{aligned} \tag{6}$$

Evaluating the cross product $\dot{\mathbf{x}} \times \mathbf{B}$,

$$\dot{\mathbf{x}} \times \mathbf{B} = \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{z} \\ \dot{r} & r\dot{\theta} & \dot{z} \\ B_r & B_\theta & B_z \end{vmatrix}, \tag{7}$$

leads to

$$\begin{aligned}\frac{d(\gamma m\dot{r})}{dt} - \gamma m r\dot{\theta}^2 &= q\left(E_r + r\dot{\theta}B_z - \dot{z}B_\theta\right),\\ \frac{1}{r}\frac{d(\gamma m r^2\dot{\theta})}{dt} &= q\left(E_\theta + \dot{z}B_r - \dot{r}B_z\right),\\ \frac{d(\gamma m\dot{z})}{dt} &= q\left(E_z + \dot{r}B_\theta - r\dot{\theta}B_r\right)\end{aligned} \tag{8}$$

Conservation of canonical angular momentum

Write the fields in terms of the scalar potential $\phi(\mathbf{x})$ and vector potential $\mathbf{A}(\mathbf{x})$:

$$\begin{aligned}\mathbf{E} &= -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} &= \nabla \times \mathbf{A}.\end{aligned} \tag{9}$$

The gradient $\nabla\phi$ in cylindrical coordinates is:

$$\nabla\phi = \left[\frac{\partial\phi}{\partial r}\right]\hat{r} + \left[\frac{1}{r}\frac{\partial\phi}{\partial\theta}\right]\hat{\theta} + \left[\frac{\partial\phi}{\partial z}\right]\hat{z}. \tag{10}$$

And the curl $\nabla \times \mathbf{A}$ is:

$$\nabla\times\mathbf{A} = \left[\frac{1}{r}\frac{\partial A_z}{\partial A_\theta} - \frac{\partial A_\theta}{\partial z}\right]\hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right]\hat{\theta} + \left[\frac{1}{r}\left(\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial\theta}\right)\right]\hat{z}. \tag{11}$$

Axial symmetry means all partial derivatives with respect to θ are zero in Eq. (10) and Eq. (**??**). This gives the following expressions for the fields in terms of the potentials in cylindrical coordinates:

Series expansion of axially symmetric fields

Consider external fields $\mathbf{E}$ and $\mathbf{B}$ with azimuthal symmetry ($\partial/\partial\theta = 0$). In the absence of charge and current densities,

$$\begin{aligned} \nabla \cdot \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= 0, \\ \nabla \times \mathbf{B} &= 0. \end{aligned} \tag{19}$$

The vanishing curl means we can express the fields as

$$\begin{aligned} \mathbf{E} &= -\nabla\phi, \\ \mathbf{B} &= -\nabla\phi_m, \end{aligned} \tag{20}$$

for scalar potentials ϕ and ϕ_m. Additionally, the vanishing divergence leads to the Laplace equation for the scalar potential:

$$\nabla^2\phi = 0. \tag{21}$$

In cylindrical coordinates, the Laplace equation reads:

Derivation of paraxial ray equation

We will now take the *paraxial* limit of small transverse velocity relative to to longitudinal velocity: $\dot{r} \ll \dot{z}$, $r\dot{\theta} \ll \dot{z}$. If $v = (v_r^2 + v_\theta^2 + v_z^2)^{1/2}$ is the total velocity, where $v_r = \dot{r}$, $v_\theta = r\dot{\theta}$, and $v_z = \dot{z}$, the paraxial limit gives $\dot{z} = (v^2 - \dot{r}^2 - r^2\dot{\theta}^2)^{1/2} \approx v$. We will therefore only keep terms linear in r in the equations of motion.

We will also add the self-generated fields to the equation of motion, writing

$$\begin{aligned} \mathbf{E} &= \mathbf{E}^{(ext)} + \mathbf{E}^{(self)}, \\ \mathbf{B} &= \mathbf{B}^{(ext)} + \mathbf{B}^{(self)}. \end{aligned} \tag{35}$$

[Is there a way to derive the ray equation first, then add the self-fields later?] To first order in r, the external fields are:

$$\begin{aligned} E_r^{(ext)} &\approx \frac{r}{2}\frac{\partial^2 V(z)}{\partial z^2}, \\ E_z^{(ext)} &\approx -\frac{\partial V(z)}{\partial z}, \\ B_r^{(ext)} &\approx -\frac{r}{2}\frac{\partial B(z)}{\partial z}, \end{aligned} \tag{36}$$

Envelope equations for axially symmetric beams

Envelope equations for axially symmetric beams

The strategy in this section will be to compute the second moments of the transverse distribution function $f(x, x', y, y', s)$. For axially symmetric beams, the Vlasov equation will generate a set of differential equations for the evolution of the second moments such as $\langle rr \rangle$, $\langle rr' \rangle$, $\langle rr'' \rangle$, etc. Combining with the paraxial ray equation will generate a set of ordinary differential equations (ODEs) for the evolution of the beam *envelope* radius $r_b = \sqrt{2\langle rr \rangle}$.

Start with the Vlasov equation:

$$\frac{\partial f}{\partial s} + \mathbf{x}' \cdot \frac{\partial f}{\partial \mathbf{x}} + \mathbf{x}'' \cdot \frac{\partial f}{\partial \mathbf{x}'} = 0, \tag{53}$$

where $\mathbf{x} = (x, y)$. Multiply the Vlasov equation by $g(\mathbf{x}, \mathbf{x}')$, an arbitrary function of the phase space coordinates, and integrate over all phase space space:

$$\iint \left[g \frac{\partial f}{\partial s} + g\mathbf{x}' \cdot \frac{\partial f}{\partial \mathbf{x}} + g\mathbf{x}'' \cdot \frac{\partial f}{\partial \mathbf{x}'} \right] d\mathbf{x} d\mathbf{x}' = 0. \tag{54}$$

Introduce bracket notation to indicate the expected value under the distribution; e.g.,

Envelope equations for elliptically symmetric beams

References